

COMAL Queue
Technical Documentation

Version 1.0.0

Comal++ / Troyanos Engineering

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List of Code Listings

1 Product and scope

1.1 Product vision

Comal++ announces ready restaurant orders using the last two digits of a manually entered ticket. A Windows PC runs the operator interface and the public display, with local announcement audio and optional multimedia. The system supports unordered kitchen output; it does not generate an order sequence.

Core operation needs neither Internet nor a network service on another machine. YouTube and weather are optional external features. Business acceptance requires the intended PC, extended TV display and speakers.

Evidence: README.md; nucleo/turnos.ts; main/main.ts

1.2 Business context

The restaurant's ticketing and sales processes remain outside Comal++. The operator calls an order when it is ready; customers read the TV and hear the announcement. The normal installation uses a single PC and an extended display connected to it.

Historical meeting notes describe an 08:30-15:00 operating day. This is context, not a schedule enforced by the application. Actual staffing, delivery procedures, noise levels and hardware must be confirmed by the restaurant.

Evidence: docs/casos-de-uso.md; docs/prueba-jornada-real.md

1.3 Implemented scope and boundaries

Implemented: input of one to six digits, display of the last two (00-99), calling and recalling, one-level correction, removal, five-minute expiry and six recent visible numbers. State is saved locally before an accepted call is announced.

Implemented: separate React operator/public windows, local audio, ducking of multimedia, import/removal of videos and banners, optional desktop YouTube, diagnostics, JSON persistence, backup/restore CLI, and migration from compatible 0.2.0 SQLite data.

The local browser transport uses HTTP and SSE on the same PC. It does not

enable LAN access, and does not support YouTube. No login, administrator account, counter assignment, delivered/cancelled order history, sales, POS integration, printing, inventory or payment processing is implemented.

Evidence: shared/contract.ts; backend/queue-store.ts; main/servidor-web.ts

1.4 Product glossary

Ticket: external order number entered by the operator. Visible number: integer 0-99, formatted as two digits. Current: most recently called visible number. Called list: up to five earlier visible numbers. Announcement: session-local ID and number sent for visual/audio playback.

Correction: one-level silent restoration of the previous call state. Removal: explicit deletion from the visible list, without business delivery semantics. Expiry: removal five minutes after the number's latest call. Delivery acknowledgement: software playback status, not proof that speakers produced sound.

Evidence: nucleo/turnos.ts; shared/contract.ts

2 Users and research

2.1 User groups

The operator enters tickets, repeats or removes numbers and manages multimedia on the local PC. The customer observes the public display without editing queue state. The installation/support owner configures Windows, audio, data backups and release acceptance.

These are operating responsibilities, not authenticated accounts. Desktop IPC restricts public-window capabilities, but the loopback HTTP API trusts local access. No separate administrative identity exists.

Evidence: vistas/operador/OperadorPage.tsx; main/seguridad-ipc.ts

2.2 Working personas and research limits

Operator persona: needs fast keyboard calling, visible confirmation and a short, actionable error when saving or audio capacity fails. Customer persona: needs legible two-digit numbers and audible announcements in the restaurant environment. Support persona: needs diagnostic routes, logs and a reversible recovery procedure.

These personas are derived from source and operational notes, not formal interviews or measured usability research. No conversion rate, satisfaction result or accessibility certification is asserted.

2.3 User scenarios

Normal call: enter 213298 and press Enter; after successful persistence, 98 becomes current and is announced. Repeat: enter 98 again; its announcement receives a new ID and its five-minute validity restarts. Correction: restore the previous call state silently once.

Saturation: six active/pending announcements prevent another call before mutation; keep the input and retry when capacity returns. Audio failure: the persisted number stays visible, while the operator checks the failed acknowledgement and repeats after fixing the cause.

Offline: local queue, voices, videos and banners continue; desktop YouTube falls back to local content. Browser YouTube is explicitly unavailable.

Evidence: backend/queue-store.ts; main/entregas-audio.ts; vistas/publica/Contenido.tsx

2.4 Daily operating journey

Before service: verify Windows extended display and output device, open Help from the header or F1, then Diagnostics; test a ticket and listen on the actual speakers. During service: call ready orders, monitor the compact notices and per-number audio status, and use repeat/remove only as appropriate.

After service: close the application before backing up the effective data directory. After recovery or upgrade: verify version, queue, content, sound and display before serving customers. Previous-day state and old announcements do not replay automatically.

Evidence: docs/user-manual/operador.md; docs/operacion-recuperacion.md

3 Requirements

3.1 Product requirements

PR-01: announce ready orders without integrating the sales system. PR-02: keep capture separate from customer presentation. PR-03: preserve local state before acknowledging a call. PR-04: operate locally without mandatory Internet. PR-05: provide support diagnostics and recoverable data.

Requirements below describe implemented behavior of 0.4.0. Historical RF/RN/CU labels remain in docs/casos-de-uso.md with an explicit current-status table; they must not be treated as a different implementation baseline.

3.2 Functional requirements

FR-01: accept one to six decimal digits and announce the last two, including 00; reject invalid input without changing the display. FR-02: maintain one current and at most five called numbers without duplicates. FR-03: recalling reorders a listed number, or repeats current audio without duplicating it.

FR-04: support silent one-level correction and explicit removal. FR-05: expire each number five minutes after its latest call, and reset on day change. FR-06: reject calls when six audio announcements are outstanding, with a warning from five.

FR-07: play notice plus selected local voice, one or two repetitions, with media ducking. FR-08: import supported media asynchronously and report skipped files without overwriting existing content. FR-09: expose diagnostics, Help, volume, theme and size controls. FR-10: backup and restore through the CLI with the app closed.

Evidence: nucleo/turnos.ts; main/contenido.ts; main/respaldo.ts

3.3 Non-functional requirements

NFR-01: normal queue and local media remain functional offline. NFR-02: atomic JSON writes publish only complete files; a save failure rejects the action. NFR-03: renderer isolation, sandbox, CSP and validated IPC enforce desktop capability boundaries. NFR-04: web transport binds only to 127.0.0.1 and validates Host and write Origin.

NFR-05: queue audio is bounded to six announcements, start wait to 45 seconds and task duration to 30 seconds. NFR-06: logs rotate at 5 MB with five previous files. NFR-07: Windows x64 packaging must pass installation, upgrade and rollback validation before release.

No measured physical audio latency, formal SLO or guaranteed power-loss durability of the storage device is asserted.

Evidence: shared/politica-anuncios.ts; main/escritura-atomica.ts; main/log.ts; .github/workflows/validate.yml

3.4 User stories

US-01: as operator, call a complete ticket quickly while only its last two digits appear publicly. US-02: as operator, repeat or correct a call without duplicated visible numbers. US-03: as operator, receive a compact notice and keep my input when audio is full or persistence fails.

US-04: as customer, see readable current/recent numbers and hear local voice. US-05: as operator, import videos/images and keep a local fallback when desktop YouTube fails. US-06: as support owner, inspect Diagnostics and recover an app-closed backup safely.

Evidence: vistas/operador/OperadorPage.tsx; vistas/publica/Publica-Page.tsx

3.5 Acceptance criteria

AC-01: 213298 becomes 98; 00 is accepted; seven digits, signs and nonnumeric input leave state unchanged. AC-02: a recall leaves unique numbers in order; repeating current creates another announcement ID. AC-03: removal clears pending correction and pending announcements for that number.

AC-04: six outstanding announcements cause a further call to fail without state/ID/undo mutation. AC-05: failed persistence produces no accepted call or new announcement. AC-06: restore rejects corrupt data without changing the target and preserves previous target data on successful replacement.

AC-07: Help is available in the header and by F1, without a redundant sidebar entry; size offers seven steps from 70% to 130%; both clocks show a 12-hour time with am/pm. These interface changes are coordinated with the parallel

implementation session.

Hardware acceptance remains separate: verify HDMI, Windows scale, speakers and real connected YouTube if used.

Evidence: main/store.test.ts; main/respaldo.test.ts; scripts/smoke-interface.mjs; scripts/smoke-theme.mjs

3.6 Requirements traceability

FR-01/02/03/04/05 -> nucleo/turnos.ts and backend/queue-store.ts -> nucleo/turnos.test.ts, main/store.test.ts, main/vigencia.test.ts, main/reinicio-diario.test.ts.

FR-06/07 -> main/entregas-audio.ts, vistas/publica/useAnuncios.ts and audio.ts -> main/entregas-audio.test.ts, nucleo/cola.test.ts, scripts/audio-lifecycle.test.mjs, smoke-hardening.mjs and smoke-audio-controls.mjs.

FR-08 -> main/contenido.ts -> main/contenido.test.ts and smoke-media.mjs. FR-09 -> operator components and common theme/size/clock -> smoke-interface.mjs, smoke-theme.mjs and main/diagnostico.test.ts. FR-10 -> main/respaldo.ts, scripts/datos.ts -> main/respaldo.test.ts and validar-recuperacion.mjs.

NFR-03/04 -> seguridad-ipc.ts, politica-csp.ts, safe-file.ts, servidor-web.ts -> corresponding main/*.test.ts and smoke-browser.mjs. Packaging -> package-desktop.mjs -> smoke-installed.mjs, smoke-upgrade.mjs and workflows. Test existence is not evidence of a passing run.

4 User experience

4.1 Interface overview

The operator workspace has Turnos and Multimedia navigation, a header with Help, voice/media volume, seven size levels (70-130%) and Azul/Morado theme. Diagnostics is accessed at the end of Help. F1 opens Help and Escape closes it.

Both views use a 12-hour clock with am/pm. The theme selector reserves width to avoid a layout shift. The coordinated wave background uses subtle continuous motion over 24-31.8 seconds, with 33 sinusoidal geometry samples per cycle; reduced-motion preferences are respected.

Theme and size are stored in localStorage and propagated by storage events to same-origin views. Desktop and HTTP browser storage are different origins: their preferences do not automatically synchronize with each other.

Evidence: vistas/comun/tema.ts; tamanio.ts; hooks/useClock.ts; vistas/operador/OperadorPage.tsx

4.2 Operator interface

Turnos accepts one to six ticket digits; Enter calls the last two. A successful call clears capture. Invalid input, audio saturation or a failed save must not be mistaken for a new call. Each visible row offers Anunciar and Quitar with confirmation; the operator can omit those confirmations for the current session.

Corregir última captura restores one previous call silently. Ctrl+Z edits the input text, not queue history. Persistent warnings are grouped in a compact expandable notice; row statuses identify En espera, Anunciando, Anunciado, Falló el audio and discarded announcements. Short feedback does not replace Diagnostics.

Help is in the header and F1, without sidebar duplication. There are no counters, delivered-state workflow or order-history screen.

Evidence: vistas/operador/OperadorPage.tsx

4.3 Public display

The public display presents one current number and up to five earlier numbers, large announcement cards, multimedia and a footer. It does not paginate an unbounded list. Numbers expire five minutes after their latest call.

DisplayCanvas fits and centers a 1920x1080 composition uniformly in available CSS pixels. Different aspect ratios leave black margins. This fit scale is separate from the 70-130% size preference. A secondary Windows display gets fullscreen; without one the public view is a normal window on the primary display.

Technical diagnostics and operator capture do not appear publicly. Software audio acknowledgements do not prove that customers heard the speakers.

Evidence: vistas/publica/PublicaPage.tsx; DisplayCanvas.tsx; main/ventanas.ts

4.4 Configuration and support interface

There is no administrator account or separate administration page. Multimedia import/removal, YouTube and volumes are operator capabilities. Diagnostics is a support panel accessed through Help and refreshed every ten seconds while open.

Additional config.json fields can be edited locally; watch-based validation keeps the previous configuration while JSON is incomplete. For backup/restore or manual file repair, close the app first and use the effective path shown in Diagnostics.

Evidence: main/config.ts; vistas/operador/DiagnosticoPanel.tsx; docs/operacion-recuperacion.md

4.5 Interaction and accessibility guidance

Preserve fast keyboard capture, focus return after Help and semantic labels on icon controls. Keep warnings concise, expandable and attached to their consequence; never label failed software audio as announced. Do not hide save failures behind success feedback.

Use the intended TV at viewing distance to assess legibility. Preserve reduced-motion behavior for backgrounds. The existence of focus styles, labels and

reduced-motion CSS does not establish an accessibility certification or screen-reader acceptance.

Historical screenshots of the 0.2.0 interface are not authoritative; use the current screenshot inventory.

Evidence: docs/user-manual/SCREENSHOTS_REQUIRED.md; vistas/comun/styles/interfaz.css

5 Architecture

5.1 System overview

Comal++ is an Electron application with two React entry points built by Vite. The main process composes queue state, JSON storage, validated IPC, local content, window lifecycle and a loopback HTTP server. No NestJS or Socket.IO server runs in the current product.

Desktop flow: React -> transport -> sandboxed preload -> validated IPC -> queue store -> pure core and JSON storage. Browser flow: React -> browser transport -> HTTP actions and SSE -> the same queue application.

The operator and public display are separate windows. IPC public permissions differ from operator permissions; HTTP is a trusted-local transport and has no per-browser identity.

Evidence: main/main.ts; main/adaptador-ipc.ts; backend/queue-store.ts; shared/contract.ts

5.2 Frontend architecture

vistas/operador/main.tsx and vistas/publica/main.tsx mount separate React 19 pages. Common provider and transport encapsulate Electron/browser differences. Components never import Electron directly for queue operations.

Common modules implement localStorage theme/size, clock, weather, icons, modal and audio controls. The public view handles the announcement FIFO, Web Audio playback, cards, media fallback and uniform display fitting.

Browser transport is provided by public/browser-transport.js. The same-origin browser preferences synchronize between browser views, but not across the desktop custom-protocol origin.

Evidence: vistas/comun/turnero.tsx; vistas/comun/transport.ts; vite.config.ts

5.3 Queue application and composition

backend/queue-store.ts is the sole queue-state owner, with injected storage/capacity/date ports. nucleo/turnos.ts contains pure state transitions.

main/store.ts binds JSON persistence and composition-specific ports. main/main.ts owns startup and shutdown.

Successful actions save state before notifying subscribers and sending an announcement. On save failure the prior queue, timestamps, undo and announcement sequence are restored. Audio capacity is checked before accepting a call. Expiry uses a single timer and retries failed persistence after five seconds.

This is a single-process local design. It has no distributed queue, service container or database repository.

Evidence: backend/queue-store.ts; main/store.ts

5.4 Data flow and action boundaries

LLAMAR -> validate input -> compute pure transition -> check audio capacity -> update latest-call time -> atomic save -> snapshot and optional announcement. Invalid capture returns CAPTURA_INVALIDA; full audio capacity or failed saving rejects before an accepted state notification.

DESHACER restores one previous snapshot without announcing; QUITAR removes a number and clears undo. Expiry and day change save before publishing. Public renderers report software playback statuses by ID and number.

Media import uses a main-process picker, asynchronous copy into a hidden destination temporary, size verification and exclusive hard-link publication. Existing files are retained with numbered names. Inventory updates reach IPC and SSE subscribers.

Evidence: backend/queue-store.ts; main/contenido.ts; main/entregas-audio.ts

5.5 Persistence and legacy database migration

The active product stores config.json, estado.json, turnero.log and editable contenido/ under the effective data directory, normally %APPDATA%/comal-local/datos in packaged 0.3.1. TURNERO_DATOS overrides it for isolated tests; development defaults to the repository root.

estado.json contains fecha, actual, llamados, optional desde timestamps and

guardadoEn. Undo history and announcement deliveries are session memory and do not survive restart. Only same-day, unexpired numbers return; old announcements never replay automatically.

SQLite is used only to read a supported 0.2.0 profile during migration. Up to six unique ready numbers last announced today are imported, with a fresh five-minute validity window; compatible messages and YouTube configuration are mapped. The legacy database remains untouched. The data/ SQLite files in the repository are legacy material, not active storage.

Evidence: main/persistencia.ts; main/carpeta-datos.ts; main/migracion-02.ts

5.6 Queue domain

State is actual (number or null), llamados (at most five earlier unique numbers), and one optional undo snapshot. There are no persistent ready/delivered/cancelled records or UUID-based orders in this version.

Input must be one to six decimal digits; normalizar takes the last two. 213298 becomes 98 and 100 becomes 00. Calling current repeats it without reordering; calling an earlier number promotes it and removes its prior occurrence. A new number drops the oldest if six were visible.

Each call refreshes five-minute expiry. Removal and expiry are silent; explicit removal discards undo. Undo cannot restore already expired numbers, and is unavailable after restarting. Day change clears queue state through persisted application logic.

Evidence: nucleo/turnos.ts; backend/queue-store.ts

5.7 IPC and local browser events

Desktop subscriptions broadcast estado, config, contenido, pantallas and entregaAudio through named IPC channels. The sandboxed preload exposes the limited TurneroApi; its inline channel literals are checked against shared/ipc-channels.ts.

Browser GET /api/inicial fetches current state; GET /api/events uses Server-Sent Events with JSON {tipo, datos}. Event types are estado, config, contenido, pantallas and entregaAudio. The browser test requests a free port, reads the

origin from its own log and checks the diagnostic data directory, preventing accidental use of another running instance on 4317. This is SSE, not Socket.IO or WebSocket. Reconnection refreshes current state; it does not replay old announcements.

Announcement IDs increase during a session. Delivery statuses are validated by ID and number; terminal states cannot be reversed by late acknowledgements. The bounded in-memory delivery history retains up to 100 entries without evicting active ones. Multiple public windows can play the same announcement: there is no single-output election.

Evidence: public/browser-transport.js; main/servidor-web.ts; main/entregas-audio.ts

5.8 Audio and multimedia architecture

Desktop source priority: configured and available YouTube, playable local videos, banners, then welcome branding. Browser skips YouTube and uses local fallback. Local videos use a shuffled bag, avoid immediate repeat when possible and skip failures. Banners default to eight seconds.

The announcement FIFO allows six tasks including active, a 45-second start deadline and 30-second execution limit. Local Web Audio plays notice and selected voice with one or two repetitions, after multimedia ducking is confirmed. Missing, muted, undecodable or interrupted voice does not confirm success.

Audio inventory prefers nonempty MP3, otherwise WAV; 00-99 and notice are required. Byte size/timestamps affect revisionAudio, invalidating decoded buffers on replacement. YouTube starts muted, confirms volume before playback and reports health from player state; failure triggers local fallback.

Evidence: vistas/publica/audio.ts; useAnuncios.ts; Contenido.tsx; YouTubeVideo.tsx; shared/politica-anuncios.ts

5.9 Trust and capability boundaries

No user login or account roles exist. Desktop authorization is based on registered window, exact page URL and main frame, with runtime payload validation. Public desktop renderers cannot dispatch queue mutations or

request full diagnostics; they have only the allowed playback, reading and reporting capabilities.

The HTTP server listens on 127.0.0.1. Every request requires the exact Host. POST additionally requires the exact local Origin, application/json and X-Turnero-Cliente: navegador. These are local request checks, not user authentication; any local client that supplies them can access the exposed API.

Do not describe the public browser UI as an authenticated security role. LAN administration, remote devices and public hosting are unsupported.

Evidence: main/seguridad-ipc.ts; main/servidor-web.ts

5.10 Implemented architectural decisions

AD-01: pure queue rules with a single application state owner and injected storage ports. AD-02: preserve installed JSON field names and use atomic writes before broadcasting accepted actions. AD-03: separate operator/public windows with a validated sandboxed preload.

AD-04: optional trusted-local HTTP/SSE transport reuses the same application logic, bound to loopback. AD-05: local prerecorded voice catalog makes normal service independent of synthesis/network availability. AD-06: bounded announcement FIFO with explicit failed/discarded acknowledgements.

AD-07: migrate legacy SQLite read-only without importing historical announcements or retaining old order semantics. AD-08: Windows NSIS installer and signed, evidence-based release gate. These are observations of implementation, not invented original decision rationales.

Evidence: docs/current-architecture.md; package.json; .github/workflows/cd.yml

6 API and contracts

6.1 API overview

The primary desktop API is window.turnero, typed as TurneroApi in shared/contract.ts. It exposes initial state, validated actions, subscriptions, volume/YouTube configuration, content management, diagnostics, logging and playback reports.

The optional HTTP interface is an internal same-PC transport at http://127.0.0.1:4317. Its routes are listed below. There is no versioned public API, NestJS controller, Socket.IO endpoint, turn-history resource or promise of external client compatibility.

TURNERO_PUERTO='0' requests a free local port for isolated tests; the actual origin is logged. With the normal default, if 4317 is busy, failure is logged and desktop windows remain usable. Browser transport needs the running app and built views.

Evidence: shared/contract.ts; main/servidor-web.ts

6.2 HTTP routes and request requirements

GET / redirects to /vistas/operador/index.html; GET /publica redirects to /vistas/publica/index.html. GET /api/inicial returns Inicial with browser content URLs. GET /api/diagnostico returns local diagnostics. GET /api/events streams JSON event envelopes.

POST /api/accion accepts LLAMAR, DESHACER or QUITAR. POST /api/volumen accepts [voz, multimedia]. POST /api/youtube accepts null only; a URL is rejected in browser mode. POST /api/salud accepts [tipo, estado]. POST /api/registro accepts a bounded string. POST /api/audio accepts [id, n, estado].

POST /api/contenido/importar accepts videos or banner and opens a native picker on the PC. POST /api/contenido/quitar accepts an inventory URL beginning /contenido/. POST /api/contenido/abrir accepts a category or null and opens the local folder.

GET /contenido/... and built static assets resolve only within allowed roots, supporting single byte ranges. POST requires Origin equal to the actual

loopback origin, Content-Type exactly application/json, and X-Turnero-Cliente exactly navegador. Host must match actual bind address and port; query strings/hashes and unsupported methods are rejected. Body text length is bounded to 65536.

Evidence: main/servidor-web.ts; main/seguridad-ipc.ts

6.3 Data contracts and compatibility

Accion = {tipo: LLAMAR, entrada: string} | {tipo: DESHACER} | {tipo: QUITAR, n: number}. Instantanea = {actual, llamados, puedeDeshacer, optional persistencia/advertenciaRecuperacion}. ResultadoDespacho = {instantanea, efecto: ANUNCIAR/CAPTURA_INVALIDA/null, anuncio}.

Anuncio = {id: number, n: number, optional limiteInicio}. EntregaAudio = {id, n, estado, fecha, optional motivo}; estado is pendiente, reproduciendo, reproducido, fallo or descartado. IDs are local to the running session and are not durable order UUIDs.

Inicial contains instantanea, config, inventario, pantallas and optional delivery state/history. Inventario contains videos/banner URL arrays, 100 voice slots, notice URL and optional revisionAudio. Pantallas.publica is tv, ventana or ninguna. Result import reports agregados, omitidos, cancelado and optional motivos.

Config keys and persisted JSON are compatibility-sensitive. See the configuration section for defaults/ranges; do not replace them with legacy SQLite settings contracts.

Evidence: shared/contract.ts; main/persistencia.ts

6.4 Error semantics

Invalid capture is a successful dispatch result with CAPTURA_INVALIDA and no mutation. Full announcement capacity and failed saving reject the call; the operator preserves its ticket for retry. The interface removes the Electron remote-method wrapper from human-facing errors and uses brief queue-full feedback.

HTTP validation/operation failures normally return 400 with {error: message}. Host or write-header mismatch returns 403. Unsupported

routes/query/method combinations return 404 or 405 according to route handling. Invalid media range returns 416. After a streamed response starts, a later error destroys the response instead of returning another JSON document.

These status codes are implementation behavior, not a REST error schema promising separate 409/422/503 domain codes.

Evidence: main/servidor-web.ts; vistas/operador/OperadorPage.tsx

7 Quality and verification

7.1 Quality strategy

Separate code evidence, executed automated checks, historical reports and target-hardware acceptance. A test script or old PASS does not certify a newly built installer. Retain exact commit, artifact SHA-256, environment and results for each delivery.

Validate pure rules and persistence in unit tests, then transport/audio/UI in Electron smoke tests, then the exact installer and upgrade fixture. The 390-minute endurance run and physical sound/display checks are separate acceptance requirements.

Evidence: docs/lista-publicacion.md; .github/workflows/validate.yml; release.yml

7.2 Test commands and scope

npm test validates the audio catalog and nucleo/main/lifecycle tests. npm run lint and npm run format:check validate maintained code/config. npm run build type-checks and compiles both renderer and main entry points.

test:desktop covers windows and local playback; test:browser covers HTTP/SSE, browser audio reports and logs; test:hardening covers announcement identity/capacity and asynchronous import; test:media covers local media. test:interface and test:theme cover controls, help, preferences and UI synchronization. test:destructive covers destructive regressions in isolated data.

test:recovery runs the actual backup/restore CLI and stores hashes/results. test:installed needs an already built installer. test:upgrade needs a verified 0.2.0 fixture. test:full-day:preflight is a short startup check; test:full-day runs 390 real minutes. Its updated preflight checks eight confirmed announcements, saturation rejection without mutation, invalid input, correction and reload without replay. The full run drains outstanding audio before corrections/reloads, checks TV/memory/file agreement, bounds capacity retries to 120 seconds and exercises recovery halfway through the real run. A preflight PASS does not establish a full-run PASS. Manual display-scale, background and audio-control smoke scripts remain available.

Evidence: package.json; scripts/; docs/prueba-jornada-real.md

The local endurance run on 2026-10-02 completed PASS after 390 real minutes: 124/124 announcements and 422 successful integrity checks. Evidence and limits: docs/resultado-jornada-2026-10-02.md. This is source-checkout evidence; final candidate identification and target hardware acceptance remain required. No prolonged video or physical audibility was validated.

8 QA evidence: executed scope and limits

8.1 Evidence boundary

This section reports only evidence present in the repository and the dated QA report. A designed scenario is not an executed scenario. The current record covers a real local Selenium run, its TestLink read-back, and infrastructure checks; it does not certify the complete product regression.

Document ID	Observed action
COMAL-QA-001	Selenium enters 42 in the operator view and the public view reads 42.
COMAL-QA-003	Selenium submits an invalid capture and verifies that state is not muta
COMAL-QA-011	Selenium corrects the last capture without announcing it.
INFRA-ADAPTER-001	Deliberate negative report is written and read back as TestLink status f

Evidence: qa/management/reporte-final.md; qa/management/alcance-y-entregable-final.md

The integration report identifies TestLink case COMAL-50, build 7, execution 9, and the negative control in execution 10. The latest UI suite created build 13 and read back executions 21, 22, and 23 as PASS. These identifiers are the observed mappings for those runs; they are not interchangeable with the documentary COMAL-QA-xxx identifiers.

8.2 TestLink and Docker infrastructure

The local TestLink stack is defined in qa/testlink/compose.yaml. It binds the web service to 127.0.0.1:8090, uses pinned image digests, and persists the database, application, logs, and upload areas in named volumes. The

repository records successful checks for login, authenticated index, XML-RPC, and HTTP blocking of `/install` and `/docs`. Recreating containers without removing volumes preserved the schema, catalog, executions, and attachment record.

Component	Repository evidence
TestLink	`qa/testlink/setup.ps1`, `verify.ps1`, XML-RPC adapter, local-only endpoint
Catalog	43 designed scenarios in `qa/catalogo/casos-prueba.json`; 12 use cases and
Persistence	MySQL and TestLink named volumes; persistence check recorded as PASS
Automation	`qa:integration` and `qa:ui`; UI report targets the three COMAL cases above

8.3 Final regression (phases 5 and 6)

All 46 catalog scenarios were executed on 8 November 2026 (04:35–04:45 UTC) with `npm run qa:regression` against Comal 1.0.0 and reported to TestLink build 26. Every result was read back from TestLink and matched.

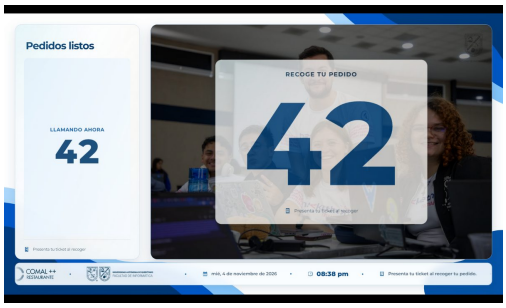
Status	Cases	Meaning
PASS	37	every automated check passed
FAIL	0	—
BLOCKED	9	automated checks passed; physical or human part pending

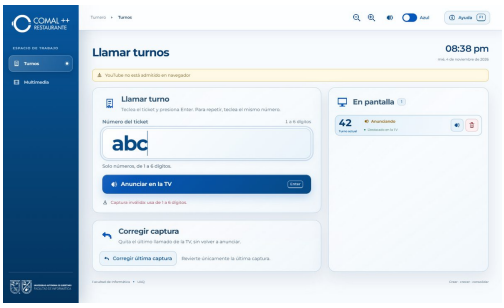
Each case is mapped in `qa/automation/regression-map.mjs` to unit and integration tests (`node:test`, 65 file/pattern mappings, each required to match at least one test), existing Electron smoke suites, 14 Selenium end-to-end scenarios and the TestLink integration smoke. A case with a manual component is never reported as passed. BLOCKED cases: 007 (latency on target hardware), 015 and 016 (audible checks), 017 and 029 (real Internet for desktop YouTube), 022 (one/two physical monitors), 023 (6.5-hour full-day run; only the preflight ran), 037 (legibility at distance) and 043 (1.0.0 installer not yet built). Per-case evidence: `docs/qa/regression-summary.json`.

8.4 Attachments and local screenshots

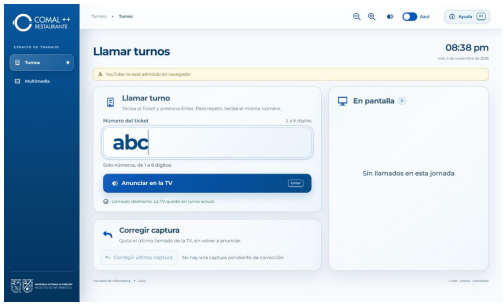
The 14 Selenium screenshots were attached to their TestLink executions as `<case>.png`. The earlier attachment error `6002` was caused by TestLink's

20-character file-name limit and is resolved (see Known limitations). Representative screenshots are kept in the documentation directory:





Execution evidence for `COMAL-QA-001` and `COMAL-QA-003`.



Execution evidence for `COMAL-QA-011`.

8.5 Pending validation and limitations

- Execute the nine BLOCKED cases on the target hardware, including the 6.5-hour `npm run test:full-day`, and build the 1.0.0 installer for `test:installed/test:upgrade`.
- Keep the TestLink image and PHP 7.4 compatibility limitation under review; the repository does not claim current upstream support.

8.6 Reproducible commands

From the repository root, the evidence documents the following PowerShell workflow. Credentials remain in the ignored local `qa/testlink/.env`.

```
.\qa\testlink\setup.ps1
.\qa\testlink\verify.ps1
npm run qa:integration
npm run qa:ui
node qa\automation\import-catalog.mjs
```

```
npm run qa:regression
```

The final regression above was produced by the last command.

8.7 Representative test cases

TC-01: valid/invalid ticket boundaries, including 00 and overlong input. TC-02: recall order, duplicate prevention, silent undo, removal and timed expiry. TC-03: persistence rejection preserves previous state; corrupt input is preserved for diagnosis; prior-day state starts empty.

TC-04: audio saturation rejects before mutation; wrong/late acknowledgements cannot overwrite delivery state; pending obsolete announcements are discarded. TC-05: import is asynchronous, concurrent names remain distinct, truncated copies fail and low space is handled.

TC-06: forged IPC sender/frame/URL and hostile local file traversal are rejected. TC-07: HTTP Host/Origin/header checks, SSE, browser diagnostics and persisted log effects. TC-08: installer first run, legacy migration, rollback and app-closed recovery hashes.

TC-09: Help header/F1, no sidebar Help, seven zoom levels, am/pm clocks, compact warnings and stable theme layout. Consult each script for assertions; presence here does not assert that this branch ran the application suite.

Evidence: main/*.test.ts; nucleo/*.test.ts; scripts/smoke-*.mjs

8.8 Known limitations

Normal operation is one Windows x64 PC; remote/LAN devices are unsupported. Browser YouTube is disabled. Desktop YouTube, weather and embedded-content availability need real Internet acceptance. There is no authenticated administrator or persistent order history.

Six visible recent numbers are not an order ledger; a seventh distinct call displaces the oldest and each number expires after five minutes. Six outstanding audio tasks may temporarily prevent further calling. Restart does not recover undo or replay old announcements.

Playback acknowledgements prove software execution, not physical audio. Multiple public windows can duplicate sound. There is no synchronized

media timeline between windows. Theme/size preferences are origin-local and outside the JSON backup.

Public release requires approved signing identity and hardware/endurance acceptance; documentation alone does not satisfy those gates.

Evidence: README.md; docs/operacion-recuperacion.md

QA tooling defect 6002 (resolved 7 November 2026). TestLink returned attachment insertion error `6002` ("Error inserting attachment on DB") for every Selenium UI screenshot, while the integration smoke attachment was accepted. Root cause: the pinned TestLink 1.9.20 image (`ghcr.io/neiesc/testlink-code:testlink_1_9_20_fixed`) sets `allowed_filenames_regexp` so that a file-name stem may contain at most 20 characters from `[a-zA-Z0-9_-]`. `publica-42.png` passed; `COMAL-QA-001-publica-42.png` did not. The same image was rejected under its long name and accepted as `COMAL-QA-011.png`. `report-ui.mjs` and `regression.mjs` now upload evidence named after the case ID. This was a QA infrastructure defect, not a Comal product defect.

Evidence: qa/testlink (container `/var/www/html/config.inc.php`); qa/management/reporte-final.md

8.9 Technical debt and evidence gaps

Open acceptance work includes the real restaurant PC/TV/audio, connected desktop YouTube and weather, final signer trust/timestamp and a completed 390-minute report on the release candidate. Resource distribution rights require owner review.

There is no single-output coordinator for multiple public audio clients, no per-browser authentication and no cross-origin synchronization of theme/size. These are design limits, not commitments to implement new features.

Historical source references, screenshots and PDF text have been refreshed in this documentation branch. Future behavior changes must update the sources and regenerate PDFs, with visual review and explicit test evidence.

The local endurance run on 2026-10-02 completed PASS after 390 real minutes: 124/124 announcements and 422 successful integrity checks. Evidence and limits: docs/resultado-jornada-2026-10-02.md. This is source-checkout evidence; final candidate identification and target hardware acceptance re-

main required. No prolonged video or physical audibility was validated.

9 Development

9.1 Setup

Use Windows and Node.js >=22.13.0 with npm. Run npm ci from the checkout, then npm run dev. The development script starts Vite at 127.0.0.1:5173 and Electron with TURNERO_DEV_URL; it does not start a NestJS process.

For built execution, run npm run build followed by npm start. Use TURNERO_DATOS to isolate experiments from production or development data. Before review run format:check, lint, npm test and build; then appropriate smoke scripts for the changed behavior.

Evidence: package.json; scripts/dev.mjs; vite.config.ts

9.2 Project structure

main/: Electron lifecycle, adapters, config, persistence, security, content and local server. backend/: queue application and ports. nucleo/: pure rules for turns, FIFO and YouTube URLs. shared/: contracts, channels and announcement limits.

vistas/: separate operator/public React entry points, shared controls and styles. public/: static assets and browser transport. contenido/: factory voices, notice, banners and optional videos. tooling/audio/: generation and verification tools. scripts/: build, packaging, recovery and regression automation.

.github/workflows/: validation, legacy fixture, signed release and endurance. docs/: current Markdown guides, technical LaTeX hierarchy, historical reports and reproducible user PDFs. data/: legacy SQLite artifacts, not active production storage.

Evidence: repository tree; package.json

9.3 Coding standards

Use TypeScript contracts at renderer/main boundaries and runtime validators for untrusted payloads. Keep pure queue transitions in nucleo and side effects behind the queue store ports. Preserve compatibility-sensitive JSON and IPC names.

ESLint covers backend/main/nucleo/shared/vistas/scripts and vite.config.ts. Prettier format and format:check target the same maintained code, workflows and package/config files; Markdown/LaTeX are reviewed separately. .gitattributes and formatting use LF.

Add focused regressions for meaningful behavior changes; use isolated data for destructive tests. Never commit signing keys, API tokens or generated operational data.

Evidence: eslint.config.js; package.json; .gitattributes; .gitignore

9.4 Git workflow and parallel changes

Use a separate branch and checkout for parallel work; do not switch the branch of a checkout being edited by another session. Coordinate file ownership and inspect the other session's committed changes before merging or publishing documentation.

This documentation branch is codex/documentacion-integral, based on f0d842a from codex/production-readiness. The parallel session owns implementation/tests; this session owns README files, docs and audio-tool documentation. Interface details are reconciled with its final code and evidence.

Review the diff and exact base before opening a PR. A documentation branch based on unmerged implementation must declare that dependency. Do not claim a new installer or passing application run from a documentation-only change.

Evidence: docs/actualizacion-documentacion-2026-10-01.md

Stable implementation snapshot d861e5f integrates the 13 final, previously uncommitted production-readiness source/test files. The remote production-readiness ref remains f0d842a. Verification and scope are recorded in docs/actualizacion-documentacion-2026-10-01.md.

9.5 Configuration fields

config.json defaults: youtubeUrl=null; repeticiones=1 (accepted 1 or 2); volumenVoz=1 (0-3); volumenMusica=0.6 (0-1); atenuacionMusica=0.15 (0-1); segundosBanner=8 (integer 3-120); pantallaPublica=null (or integer display

ID); recargaDiaria=04:00 (24-hour HH:MM); mensajes contains two default messages.

mensajes accepts at most ten nonblank strings, each at most 160 characters. Invalid or unknown fields are logged and defaulted/ignored. Incomplete JSON during watch keeps the prior config; valid file updates propagate after a 300 ms debounce. Voice above 100% may distort.

Daily reload time remains a 24-hour configuration field even though both visible clocks use 12 hours with am/pm. Theme and size are localStorage keys comal.tema and comal.tamanio, not config.json fields and not covered by the data backup.

Evidence: main/config.ts; vistas/comun/tema.ts; tamanio.ts

9.6 Environment variables

TURNERO_DATOS: explicit data directory for isolated runs. TURNERO_PUERTO='0': request an operating-system-selected free loopback port for isolated tests; other values retain the normal default. Browser and full-day smoke scripts request isolated ports; the browser test reads its own instance's logged origin and verifies its data directory. This does not enable LAN binding. TURNERO_DEV_URL: development view origin supplied by scripts/dev.mjs; not a production LAN setting. TURNERO_SMOKE_QUICK and TURNERO_MEDIA_MODE are script-specific test options.

WIN_CSC_LINK and WIN_CSC_KEY_PASSWORD: release signing certificate and password; CSC_LINK/CSC_KEY_PASSWORD are supported packager fallbacks. COMAL_SIGNING_CERT_SHA1 selects the local certificate for –firma-local. COMAL_SIGNER_SHA1 is the approved signer thumbprint checked by CD, a different variable.

AZURE_SPEECH_KEY, AZURE_SPEECH_REGION and AZURE_SPEECH_ENDPOINT belong only to audio generation; ELEVENLABS_API_KEY belongs to its generator. They are unnecessary for playback or restaurant operation. Keep all credentials out of the repo and logs.

Evidence: scripts/package-desktop.mjs; main/main.ts; tooling/audio/generate-*.mjs; .github/workflows/cd.yml

10 Operations

10.1 Deployment

Build the Windows x64 NSIS installer with npm run desktop:build for unsigned local validation, desktop:build:signed when an approved identity is available, or desktop:build:local-signed for the documented local-certificate workflow. Outputs go to release/.

NSIS permits choosing the installation directory, uses per-user installation, creates a Start menu shortcut and does not automatically create a desktop shortcut. Factory contenido is packaged beside the app and seeded into the data directory on first startup; missing/empty required voice/notice files can be repaired from factory assets on later startups.

The effective path in Diagnostics is authoritative. For 0.3.1 it normally resides in the application profile, outside Documentos/OneDrive. Installation from USB needs the exact installer, SHA-256 and reviewed signature/acceptance status.

Evidence: package.json; main/carpeta-datos.ts; README-INSTALACION-USB.md

10.2 Production acceptance

Validate a release candidate on the intended standard Windows account with TV in extended mode and the correct sound output. Exercise disconnect/reconnect, offline content, corruption/recovery in isolated data and the exact installer upgrade/rollback.

The endurance script runs 390 minutes (6 h 30 min), matching the historical 08:30-15:00 span. PREFLIGHT_PASS is not full PASS. Logs, metrics and report must identify the artifact/environment; physical audibility still needs human acceptance.

Configure production environment reviewers and tag restrictions in GitHub, approved signing secrets/thumbprint and resource distribution rights. Declaring an environment in YAML does not prove protections are enabled.

Evidence: docs/lista-publicacion.md; docs/prueba-jornada-real.md; .github/workflows/cd.yml

The local endurance run on 2026-10-02 completed PASS after 390 real minutes: 124/124 announcements and 422 successful integrity checks. Evidence and limits: docs/resultado-jornada-2026-10-02.md. This is source-checkout evidence; final candidate identification and target hardware acceptance remain required. No prolonged video or physical audibility was validated.

10.3 Diagnostics and logs

Help -> Diagnostics displays app/Electron/Node versions, effective data/config/state/log paths, operator/public status, persistence, last successful save and errors, audio/YouTube health, media counts, visible queue length and free space. The open panel refreshes every ten seconds; individual probe failures return remaining data.

turnero.log stores JSON lines with ISO/local time, level, component, version and session. It rotates at 5 MB with .1 through .5 retained; identical messages within thirty seconds are summarized. Failed logging does not stop the application.

Public renderers report software audio/media health. Browser reports and messages use HTTP and reach the same log. There is no external monitoring service, alerting SLA or automated physical sound detector.

Evidence: main/diagnostico.ts; main/log.ts; vistas/operador/DiagnosticoPanel.tsx

10.4 Backup, restore and migration

Close Comal++ and verify its processes are stopped. Run npx tsx scripts/datos.ts backup DATA BACKUPS. The tool creates a unique dated backup of config/state/content and logs, excluding temporaries; the finished name is published only after copy completion.

Restore with npx tsx scripts/datos.ts restore BACKUP DATA –app-cerrada. The flag confirms the operator checked shutdown; it does not reliably detect all app processes. JSON, configuration keys, state dates and overlapping paths are validated before replacement. Prior target data is kept at datos.antes-de-restaurar-*.

Corrupt startup state is copied to estado.json.corrupto-* when possible and starts empty with warning; unreadable state due to I/O errors stops startup.

Old-day state never reopens prior turns. A same-day restore uses stored timestamps and does not replay audio.

Migration from 0.2.0 reads legacy SQLite unchanged; migration from Documentos to 0.3.1 copies once and retains the original. Rollback requires the old verified installer and its own pre-upgrade data backup; post-migration calls are not reverse-converted to SQLite.

Evidence: docs/operacion-recuperacion.md; main/respaldo.ts; main/migracion-02.ts

10.5 Security and release controls

Electron windows use sandbox and context isolation; mainframe/sender/page authorization and runtime payload checks guard IPC. Production CSP disallows inline scripts and unsafe-eval, restricts external origins and blocks objects/base URLs. Inline styles remain required for dynamic UI. Window navigation and popup creation are restricted.

Custom protocol and HTTP content resolution enforce allowed roots and reject escaping paths/symlinks. HTTP listens only on loopback and verifies local write headers. There is no LAN access, user identity or confidentiality boundary against a local process with data access.

CI audits high/critical dependency vulnerabilities. CD requires a matching version tag, production environment, both signing secrets and approved COMAL_SIGNER_SHA1. It checks valid signatures, exact thumbprint, code-signing use and timestamps on Setup, installed app and uninstaller, and validates the exact artifact hash before release.

Evidence: main/politica-csp.ts; main/seguridad-ipc.ts; main/safe-file.ts; .github/workflows/cd.yml

11 Status and planning

11.1 Current baseline

Package version is 0.4.0. Current architecture is local Electron/React with JSON persistence, validated IPC and HTTP/SSE loopback. The documentation baseline starts at f0d842a and includes the coordinated interface changes in the parallel session.

Implemented functionality and available tests are documented separately from their executed results. Dated September/October reports remain evidence of their own runs. The current publication status requires an approved identity and operational acceptance; this documentation update does not publish a release.

Evidence: package.json; docs/actualizacion-documentacion-2026-10-01.md; docs/verificacion-produccion-2026-10-01.md

Stable implementation snapshot d861e5f integrates the 13 final, previously uncommitted production-readiness source/test files. The remote production-readiness ref remains f0d842a. Verification and scope are recorded in docs/actualizacion-documentacion-2026-10-01.md.

11.2 Product roadmap

Recommended acceptance priorities are target-hardware legibility/audibility, clear operator recovery guidance and validation of the six-number/six-audio limits at real peak load. These are recommendations, not promised release dates or stakeholder-approved scope.

If multiple terminals, remote displays, historical order tracking or account roles are requested, define requirements and trust boundaries before extending the current local product. No such capability is implied by the optional browser transport.

11.3 Technical roadmap

Before publication, consolidate implementation commits, rerun relevant regressions and validate the exact signed candidate. Preserve CLI recovery hashes, installation/upgrade results and 390-minute evidence.

Potential future work includes single-output audio coordination and stronger local-browser identity if deployment scope changes. Upgrading major dependencies requires compatibility and installer review. These recommendations are not implemented commitments.

11.4 Release process

CI invokes reusable validation: formatting, lint, unit/audio/recovery checks and audit, then Windows build, desktop/browser/hardening/media/audio-control/interface/theme smoke, preflight, unsigned installer, installed validation and upgrade/rollback. The pinned legacy installer is built separately and its transferred SHA-256 verified.

CD runs validation on version tags and publishes only a matching v plus package version. It builds a signed candidate in production, verifies Setup/app/uninstaller trust and timestamp, checks the installed-test hash and repeats upgrade/rollback for that artifact. It uploads SHA256.txt and a CycloneDX dependency SBOM, refusing to overwrite existing releases.

A branch-only manual CD run validates without publishing. release.yml is a separate manual 390-minute self-hosted Windows endurance workflow, not the publication job.

Evidence: .github/workflows/ci.yml; validate.yml; legacy-installer.yml; cd.yml; release.yml

11.5 Metrics and interpretation

Operational evidence includes call/repeat/correction/expiry counts, software audio acknowledgements, dispatch latencies, process CPU/memory, log sizes and endurance duration. The endurance report records environment, limitations and RUNNING/PASS/FAIL or PREFLIGHT_PASS status.

Dispatch latency is not time until audible sound. Queue occupancy summaries are action-based, not time-weighted. Temperature probing is optional. No benchmark, sales metric, throughput SLO, usability score or production uptime is inferred from these reports.

Evidence: scripts/smoke-full-day.mjs; docs/prueba-jornada-real.md

12 Reference

12.1 Terminology and superseded concepts

Current terms: operador, pantalla pública, actual, llamados, instantanea, anuncio, entregaAudio, config.json and estado.json. HTTP uses SSE; desktop uses IPC. Version 0.4.0 has local JSON persistence and optional loopback browsing.

Legacy 0.2.0 terms such as NestJS, Socket.IO, turns/requests tables, counters, delivered/cancelled records and UUID order identities describe the old product. They apply only to explicit history and migration fixtures, not present operation.

The current visible range includes 00. Default repetition is one, not the historical proposal of two. Six visible recent numbers must not be confused with the separate six-task announcement capacity.

12.2 Repository map and source ownership

Use README.md and docs/README.md to navigate current guides. Architecture: current-architecture.md. Operator: user-manual/operador.md. Recovery: operacion-recuperacion.md. Media: user-manual/multimedia-requisitos.md and anuncios-e-importacion.md. Delivery: CI-CD.md, seguridad-y-publicacion.md and lista-publicacion.md.

The technical-documentation.tex master includes this 57-section hierarchy. user-manual/source/build_user_manuals.py regenerates the user and quick PDFs from the current operator guide. source/screenshots retains dated/reference captures and must not be passed off as current UI.

Dated audits, publication notes and test reports retain their historical results with status notices linking to current guides. Third-party icon README files retain their attribution; they are not product operating manuals.

12.3 Documentation status and confidence

Refreshed on 1 October 2026 against package 0.4.0, f0d842a and coordinated interface changes. All 57 technical sections were rewritten to remove the obsolete 0.2.0 runtime description. Current Markdown guides and user PDF

sources are reconciled with code; historical reports keep their original run evidence.

Code-backed facts: queue transitions, JSON fields/defaults, IPC/HTTP routes, audio policy, file limits and workflows. Acceptance still needed: target hardware, physical sound, connected desktop YouTube/weather, production protections/signing and resource distribution rights.

Validation performed for this documentation update is recorded in actualizacion-documentacion-2026-10-01.md. PDF generation/visual review and LaTeX compilation are distinct checks; do not infer a TeX compiler PASS from a generated PDF or from the previous September build.

Stable implementation snapshot d861e5f integrates the 13 final, previously uncommitted production-readiness source/test files. The remote production-readiness ref remains f0d842a. Verification and scope are recorded in docs/actualizacion-documentacion-2026-10-01.md.